Embracing Hierarchical Classification: A Multifeature Space Hierarchical Network for Hyperspectral Image Classification

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***Abstract*—Classes in hyperspectral images (HSIs) often exhibit inherent hierarchical structures, such as the familygenusspecies hierarchy in tree species. Previous studies have shown that modeling these hierarchical structures can improve classification performance. However, existing deep learning methods often overlook such structures, limiting their ability to distinguish fine-grained categories. To address this issue, we propose a multi-feature space hierarchical network (MFS-HiNet), which enhances fine-grained discrimination by modeling hierarchical relationships and guiding classification top-down. The framework consists of three key components: hierarchical structure mining (HSM), multifeature space classification network (MFSCN), and parameter inheritance strategy (PIS). Specifically, HSM automatically mines latent hierarchical relationships in HSI and constructs the hierarchy; MFSCN integrates multifeature space information for node classification to improve the discrimination of subtle interclass differences; and PIS leverages parent node parameters to guide the learning of child nodes, fully exploiting interlevel correlations. Experimental results on five HSI datasets demonstrate that MFS-HiNet outperforms existing methods in classification accuracy, validating the effectiveness of the framework and highlighting the potential of hierarchical classification. The source code will be made available at https://github.com/jin-yaoWHU/MFS-HiNet**

***Index Terms*—Hierarchical classification, hyperbolic geometry, hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL images (HSIs) are a cutting-edge remote sensing technology that allows for the precise acquisition of geospatial information across continuous spectral bands, typically ranging from visible light to near-infrared [1], [2]. Compared to traditional multispectral imaging (MSI), HSI provides more detailed and rich spectral characteristics. Furthermore, HSI possesses the unique "spatial-spectral integration" feature, enabling the extraction of both spatial and spectral characteristics of the objects, thereby facilitating fine-grained classification [3]. These advantages make HSI widely applicable in various fields, such as agricultural monitoring, environmental assessment, and geological exploration [4], [5], [6].

HSI classification is a critical step in the aforementioned applications, which aim to automatically identify land cover classes based on HSI [7]. Traditional HSI classification methods primarily rely on spectral feature analysis and machine learning techniques, such as support vector machines (SVMs) and random forests (RFs) [8], [9]. However, these approaches heavily depend on handcrafted features designed by experts, which often fail to capture the deep and complex information inherent in HSI, thereby limiting classification performance [10]. In recent years, deep learning techniques have garnered significant attention and achieved remarkable progress in HSI classification due to their powerful feature representation capabilities. Unlike traditional methods, deep learning can automatically extract high-level spectral-spatial features, thereby improving classification accuracy and demonstrating superior generalization ability in complex scenarios [11].

It is worth noting that the most state-of-the-art deep learning-based classification methods adopt a flat classification strategy, treating all classes as being at the same level and performing multiclass classification directly. However, HSI classification typically involves the identification of fine-grained land cover types, which often exhibit a natural hierarchical structure in the real world [12]. For example, in land cover classification tasks, water bodies can be further classified into lakes and rivers, while vegetation classes can be gradually refined based on biological taxonomy, progressing from broad classes to specific vegetation types [13]. However, flat classification methods are difficult to fully capture the hierarchical relationships between classes, limiting the model's generalization ability and preventing further improvements in classification accuracy [14]. To verify the potential hierarchical structure present in HSI, we performed experiments on Pavia University dataset using a convolutional neural network (CNN) model employing a flat classification approach, followed by an analysis of the confusion matrix. As illustrated in Fig. 1, there is a notable degree of confusion between Classes 3 and 8, with misclassification rates of 22.42% and 8.12%, respectively, as well as significant confusion observed between Classes 1 and 7. These findings indicate that the model encounters difficulties in discriminating between certain classes, which is likely attributable to their high spectral similarity within the hyperspectral feature space. Furthermore, as illustrated by

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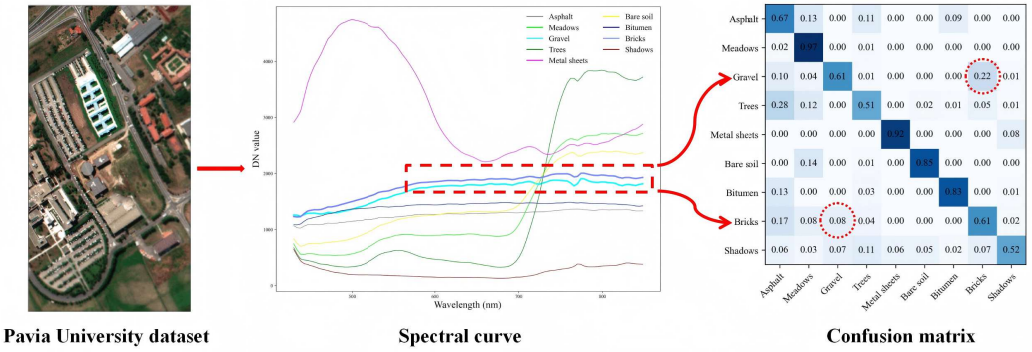


Fig. 1. Analysis of confusion matrix and spectral similarity among easily confused classes in Pavia University dataset.

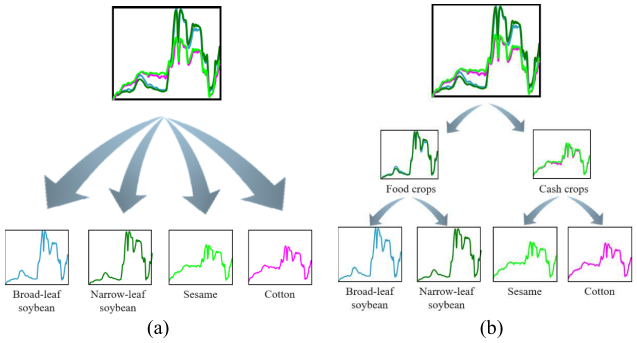


Fig. 2. Schematic of flat classification and hierarchical classification. (a) Flat classification. (b) Hierarchical classification.

the spectral curves in Fig. 1, these easily confused classes exhibit nearly overlapping spectral signatures across the entire wavelength range, showing highly consistent variation patterns. This phenomenon not only highlights varying levels of interclass similarity but also provides empirical evidence supporting the existence of an inherent hierarchical structure in HSI.

As shown in Fig. 2, unlike flat classification approaches, hierarchical classification exploits the hierarchical relationships among classes, progressively refining coarse-grained categories into fine-grained ones. By decomposing complex multiclass classification tasks into a series of subtasks, hierarchical classification can better capture the hierarchical dependencies between classes, thus enhancing classification performance. Although research on hierarchical classification in deep learning-based methods remains relatively limited, numerous studies in machine learning-based classification have demonstrated the advantages of hierarchical classification. For instance, Kumar et al. [15] proposed a hierarchical multiclassifier approach that improved classification accuracy by 4%–10% compared to conventional methods in a land cover classification task involving 12 different land cover types. Gong et al. [16] developed a hierarchical classification model based on decision trees to address the need for fine-grained vegetation classification. Compared to traditional single-classifier approaches, this method effectively reduced classification errors and was better suited for vegetation survey tasks. Similarly, Clark et al. [17] employed a hierarchical classification framework based on SVM in the San Francisco Bay Area, achieving accurate classification of forest types. Neuenschwander et al. [18] applied hierarchical classification to wetland vegetation in the western Kennedy Space Center (KSC) and Indian River wetlands, demonstrating superior performance over flat classification approaches. Additionally, Tang applied hierarchical classification to distinguish trees, shrubs, and herbaceous vegetation in the abandoned land of the eastern Haidaigou mining area in Inner Mongolia, China, achieving a final classification accuracy of 87.45%, which was 10.7%–22.7% higher than that of other classification methods [19]. These studies collectively highlight the superiority of hierarchical classification over flat classification in HSI classification tasks, offering improved accuracy and better differentiation capabilities. However, existing research has primarily focused on hierarchical classification within machine learning frameworks, while the potential of deep learning for hierarchical classification in HSI analysis remains unexplored. Therefore, we introduce hierarchical classification into deep learning-based HSI classification to fully leverage hierarchical relationships among classes, thereby constructing a more robust and efficient classification framework that enhances the accuracy and reliability of HSI analysis.

Furthermore, HSI classification methods heavily rely on computations within the Euclidean space. Although Euclidean space provides an intuitive and fundamental manifold structure, its inherent properties are not well suited for all types of data [20]. In particular, within hierarchical structures, the number of nodes increases exponentially with depth, expanding from a small number of coarse-grained classes to a large number of fine-grained ones. However, in Euclidean space, the growth of spatial volume follows a polynomial rate, which is mismatched with the exponential expansion of hierarchical structures. This discrepancy makes it challenging to achieve low-distortion embedding of hierarchical information in Euclidean space [21]. Therefore, there is an urgent need to explore alternative geometries that better capture the characteristics of hierarchical data and enhance its representational capacity. With the advancement of representation learning, studies have demonstrated that hyperbolic space, owing to its exponential distance metric, naturally aligns with the exponential expansion pattern of hierarchical structures. This property enables more effective modeling and embedding of hierarchical data, establishing hyperbolic space as a promising choice for processing hierarchical structures [22].

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Based on the above discussion, we propose a multi-feature space hierarchical network (MFS-HiNet) to automatically construct the hierarchical structure of land cover features and accurately classify the detailed land cover types. First, the features of the input original HSI data are extracted and projected into the hyperbolic space, where the hierarchical features of HSI data are explored, and the hierarchical structure is automatically constructed. Based on this structure, a hierarchical classification framework is constructed, in which a multifeature space classification network (MFSCN) is trained for each node, with model parameters propagated from parent to child nodes to enforce hierarchical consistency. Finally, the classification results of the leaf nodes are integrated to obtain the final classification result.

The main contributions of this article are as follows.

1) To address the absence of predefined hierarchical structures in HSI, we propose a hierarchical structure mining (HSM) module. This module precisely extracts hierarchical features in hyperbolic space and computes interclass distances based on these features. By transforming the distance relationships into a tree-structured hierarchy, HSM provides a structural prior for subsequent hierarchical classification.
2) Aiming to enhance the capacity of node classifiers in recognizing complex data and capturing subtle distinctions, we propose the MFSCN. This network adopts a multibranch architecture, where each branch corresponds to a distinct feature space, and integrates their outputs to improve classification accuracy and model generalization.
3) To overcome the limited information transfer inherent in top-down hierarchical classification, we design a parameter inheritance strategy (PIS). By leveraging parent-node parameters to guide the learning of child nodes, this strategy enables effective knowledge sharing, strengthens feature extraction, and improves the robustness of hierarchical classification.

This article is organized as follows. Section II reviews related works, including deep learning for HSI classification, hierarchical classification, hyperbolic geometry, and hyperbolic model for remote sensing. Section III provides a detailed description of the proposed MFS-HiNet method. Section IV presents experimental results demonstrating the effectiveness of MFS-HiNet on five real HSI datasets. Finally, Section V offers concluding remarks and outlines directions for future research.

II. Related Works

A. Deep Learning for HSI Classification

Due to the high dimensionality of HSI and the strong correlation between adjacent spectral bands, a large amount of redundant information is present in the HSI. This redundancy makes it challenging for traditional classification methods to effectively extract the spatial-spectral features embedded in HSI, thereby limiting its performance in fine-grained classification tasks [23], [24], [25]. In recent years, with the significant advantages of deep learning in signal processing and computer vision, deep learning classification methods have become a research hotspot in HSI classification [26], [27], [28].

CNNs use convolutional modules to simultaneously extract spectral and spatial features from HSI, demonstrating outstanding advantages in capturing local details [29]. As a result, CNN-based methods have rapidly become the dominant approach in HSI classification [30], [31], [32]. Xu et al. [33] proposed a spectral-spatial fully convolutional network (SSFCN) to address the computational redundancy in HSI classification caused by overlapping image patches. To address the limitation of local information reliance in existing HSI classification methods, Wang et al. [34] proposed a fully contextual network (FullyContNets) [34]. Han et al. [35] proposed a subpixel spectral variability network (S^2VNet) that integrates spectral variability and nonlinear mixture modeling, significantly enhancing HSI classification performance through subpixel information modeling and multifeature fusion.

However, CNN models are inherently constrained by their limited receptive fields, which limit their ability to model long-range dependencies [36]. In contrast, Transformer-based models have exhibited superior global modeling capabilities, leading to a surge of Transformer-based approaches in HSI classification [10], [37], [38]. Hong et al. [39] proposed SpectralFormer, a novel Transformer-based backbone network, to address the limitations of ViT in HSI processing. Yang et al. [40] proposed an interactive learning framework for ViT, termed the center-to-surrounding interactive learning (CSIL) Framework, to address key challenges in HSI classification, including geometric constraints, ambiguity in central pixel contributions, and the lack of interaction between local and distant contextual information. Feng et al. [41] proposes fractional Fourier-enhanced fusion network (FrFENet), which employs weighted fractional Fourier fusion and Pareto optimization to balance HSI and LiDAR features, enhancing classification performance while alleviating modality conflicts.

B. Hierarchical Classification

Hierarchical classification assumes that classes are not mutually independent but are organized in a tree or directed acyclic graph (DAG) structure. Within this hierarchy, each class node inherits or is contained by its parent class. The classification process explicitly leverages this structure to perform a top-down, stepwise decision-making procedure, whereby the sample's assignment to a coarse-level class is determined first, followed by progressive refinement into intermediate and fine-grained classes under the constraints of the higher-level predictions. Generally, hierarchical classification methods can be divided into local classifier approaches and global classifier approaches [42].

1) Local Classifier Approaches: Local classification approaches decompose the hierarchical classification task into a series of localized subtasks by constructing individual classifiers at different positions within the class hierarchy. Each classifier operates solely within its corresponding local label space, focusing on distinguishing between a subset of related categories. The outputs of these local classifiers are then progressively integrated to form a complete classification

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path from the root to a leaf node. Based on how the local label spaces are defined, local classification approaches are typically categorized into the following three types.

a) Local classifier per Node (LCN): In the LCN approach, a separate classifier is trained for each node in the hierarchy to determine whether a sample belongs to the category represented by that node [43]. This method offers clear structural interpretability and is amenable to parallel training. However, as the hierarchy grows deeper or contains a large number of nodes, the total number of classifiers increases substantially, resulting in high computational and maintenance costs.

b) Local classifier per Parent Node (LCPN): The LCPN method constructs a multiclass classifier for each parent node to distinguish among its immediate child classes [44]. Compared to LCN, this approach significantly reduces the number of classifiers and better leverages the contextual information within the local label space of each parent. While it strikes a balance between efficiency and accuracy, its performance may degrade in cases of class imbalance or when sibling categories exhibit high semantic similarity.

c) Local classifier per Level (LCL): The LCL strategy trains a single classifier for each level of the hierarchy to classify among all classes at that level [45]. This approach results in a simplified model architecture and facilitates the integration of intralevel information. However, it does not explicitly model dependencies across levels, which may limit its effectiveness in capturing complex hierarchical relationships.

2) Global Classifier Approaches: Global classifier approaches aim to build a unified model that directly predicts across the entire hierarchical class structure. Unlike local classifier methods that decompose the task into multiple subproblems, global classifier approaches consider all class nodes and their hierarchical relationships simultaneously during both training and inference, enabling an end-to-end holistic modeling strategy [46]. Compared to local classifier approaches, global classifier approaches offer advantages such as a more compact model architecture, explicit modeling of dependencies among classes, and stronger consistency of predictions within the hierarchical structure.

C. Hyperbolic Geometry

Hyperbolic geometry, also known as Lobachevsky geometry, is a type of non-Euclidean geometry with negative curvature. Due to the exponential volume growth characteristic of hyperbolic space, its node growth pattern is highly similar to that of tree structures. Therefore, hyperbolic space can efficiently embed tree-like structures and preserve hierarchical information in low-dimensional spaces. Currently, several different data models are used to represent hyperbolic space, including the Poincaré model, the Lorentz model, the Klein model, and the upper half-space model. The following will introduce two of the most commonly used models: the Poincaré model and the Lorentz model.

1) Poincaré Model: The Poincaré model is defined by an open ball with a radius of $-1/\kappa(\kappa < 0)$, and it can be used to describe an n-dimensional hyperbolic space with curvature κ

$$P_\kappa^n = \left\{ x \in \mathbb{R}^n : \|x\|^2 < -\frac{1}{\kappa} \right\}. \tag{1}$$

Given two points $x, y \in P_\kappa^n - 1/\kappa(\kappa < 0)$, their distance is computed as follows:

$$d_{P_\kappa^n}(x,y) = \frac{1}{\sqrt{-\kappa}} \cosh_\kappa^{-1}\left(1 - \frac{2\kappa\|x-y\|^2}{\left(1+\kappa\|x\|^2\right)\left(1+\kappa\|y\|^2\right)}\right). \tag{2}$$

The data operations in the Poincaré ball rely on the algebraic foundation provided by the gyrovector space. The exponential map enables the projection of vectors from Euclidean space into hyperbolic space, thereby realizing a nonlinear transformation from Euclidean space to hyperbolic space. Conversely, the logarithmic map serves as the inverse, mapping points from hyperbolic space back to their corresponding Euclidean space. Together, these two maps constitute the fundamental mechanism that relates points in hyperbolic geometry to vectors in the tangent space. For points $x, y \in P_\kappa^n$ and a point $v \in T_x P_\kappa^n$ in the tangent space, given that $v \neq 0$ and $y \neq x$, the exponential map $\exp_x^\kappa : T_x P_\kappa^n \to P_\kappa^n$ and the logarithmic map $\log_x^\kappa : P_\kappa^n \to T_x P_\kappa^n$ are defined as follows:

$$\exp_x^\kappa(v) = x \oplus_\kappa \left(\tanh\left(\sqrt{-\kappa}\frac{\lambda_x^\kappa\|v\|}{2}\right)\frac{v}{\sqrt{-\kappa}\|v\|}\right) \tag{3}$$

$$\log_x^\kappa(y) = \frac{2}{\sqrt{-\kappa}\lambda_x^\kappa}\tanh^{-1}\left(\sqrt{-\kappa}\|-x\oplus_\kappa y\|\right)\frac{-x\oplus_\kappa y}{\|-x\oplus_\kappa y\|} \tag{4}$$

where $\lambda_x^\kappa = 2/(1+\kappa\|x\|^2)$, and $\oplus_\kappa$ represents Möbius addition. For points $x, y \in P_\kappa^n$, the Möbius addition satisfies the following:

$$x \oplus_\kappa y = \frac{\left(1 - 2\kappa\langle x,y\rangle - \kappa\|y\|^2\right)x + \left(1+\kappa\|x\|^2\right)y}{1 - 2\kappa\langle x,y\rangle + \kappa^2\|x\|^2\|y\|^2}. \tag{5}$$

2) Lorentz Model: The Lorentz model is also known as the Minkowski model. For the Lorentz model, given two points $x, y \in R^n$, where R^n represents the n-dimensional Euclidean space, the definition of Lorentzian scalar product is

$$\langle x,y\rangle_L := -x_0 y_0 + \sum_{i=1}^{n} x_i y_i. \tag{6}$$

We use H_κ^n to represent n-dimensional hyperbolic manifolds with κ curvature

$$H_\kappa^n := \left\{ x \in R^{n+1} : \langle x,x\rangle_L = 1/\kappa, x_0 > 0 \right\}. \tag{7}$$

And for $x, y \in H_\kappa^n$, Lorentzian scalar product satisfies

$$\langle x,y\rangle_L \leq 1/\kappa, \text{ and } \langle x,y\rangle_L = \kappa, \quad \text{when } x = y. \tag{8}$$

Given two points $x, y \in H_\kappa^n$, their distance is computed as follows:

$$d_{H_\kappa^n}(x,y) = \frac{1}{\sqrt{-\kappa}}\cosh_\kappa^{-1}\left(\kappa\langle x,y\rangle_L\right). \tag{9}$$

For points $x, y \in H_\kappa^n$ and a point $v \in T_x H_\kappa^n$ in the tangent space, the exponential map $\exp_x^\kappa : T_x H_\kappa^n \to H_\kappa^n$ and the logarithmic map $\log_x^\kappa : H_\kappa^n \to T_x H_\kappa^n$ are defined as follows:

$$\exp_x^\kappa(v) = \cosh\left(\sqrt{-\kappa}\|v\|_L\right)x + \sinh\left(\sqrt{-\kappa}\|v\|_L\right)\frac{v}{\sqrt{-\kappa}\|v\|_L} \tag{10}$$

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$$\log_x^{\kappa}(y) = \frac{\cosh^{-1}(\kappa\langle x, y\rangle_L)}{\sinh\left(\cosh^{-1}(\kappa\langle x, y\rangle_L)\right)}(y - \kappa\langle x, y\rangle_L) \tag{11}$$

where $\|v\|_L = \sqrt{\langle v, v\rangle_L}$ represent the Lorentz mode of v.

D. Hyperbolic Model for Remote Sensing

Since real-world data generally exhibit an inherent hierarchical structure, hyperbolic deep neural networks have demonstrated broad application prospects across various research fields. Domains such as natural language processing, knowledge graph completion, recommender systems, computer vision, and computational biology have all benefited from these advancements [47]. As research on hyperbolic geometry in deep learning and machine learning continues to progress, its applications in the field of remote sensing are also gaining attention, revealing significant potential for further development.

Sun et al. [48] proposed a novel hyperbolic clustering band hierarchy (HCBH) method, which utilizes hyperbolic clustering to construct a band hierarchy based on the similarity of bands in hyperbolic space. This method aims to select an informative set of bands for HSI analysis [48]. Li et al. [49] introduced a hybrid attention-enhanced neural network, which employs a similarity-based hybrid attention module constructed in both Euclidean and hyperbolic spaces to accomplish the semantic segmentation of remote sensing images. Sun et al. [50] proposed a hyperbolic space-based autoencoder (HSAE), which embeds HSIs into hyperbolic space and supervises the training of hidden representations to preserve more hierarchical features. Finally, an anomaly detector based on hyperbolic distance is used to identify anomalies in a more robust manner [50]. Hamzaoui et al. [51] employed a hyperbolic prototype network as a meta-learning method, embedding scene images into hyperbolic space to tackle the few-shot remote sensing scene classification task. Experimental results demonstrate that hyperbolic embeddings outperform Euclidean embeddings in representational capacity, making computation in hyperbolic space a promising alternative for remote sensing applications [51]. Cui et al. [52] leveraged the exponential spatial expansion property of hyperbolic space to design a graph-embedding model for remote sensing scenes. This model effectively integrates visual features, hierarchical structures, and intrinsic structures of remote sensing objects within a finite-dimensional space, significantly enhancing its ability to represent complex remote sensing scenes with multiple structures [52].

III. Proposed Method

In this section, the proposed MFS-HiNet will be introduced, including backbone, HSM, MFSCN, and PIS. As shown in Fig. 3, MFS-HiNet adopts the LCPN strategy to construct its overall framework. The detailed processing pipeline is described as follows: First, the backbone is used to extract features from the HSI, which are then mapped into the hyperbolic space to explore hierarchical representations and automatically construct the hierarchical structure. Based on the hierarchical structure, a hierarchical classification framework is designed, where a MFSCN is trained for each node, and PIS is utilized to transfer model parameters from parent nodes to child nodes. Finally, the classification outcomes of the leaf nodes are aggregated to derive the final classification decision.

A. Backbone

In the MFS-HiNet, we use a simple CNN-based network as the backbone, which is used for feature extraction within the HSM modules and MFSCN.

Specifically, the backbone structure is designed as follows. The backbone incorporates 1×1 convolutions to eliminate redundant spectral data and achieve dimensionality reduction, thereby mitigating the Hughes phenomenon caused by the high similarity between adjacent bands in HSI. Additionally, to enhance the capability of the backbone in extracting spatial-spectral features from HSI, the backbone utilizes depthwise separable convolutions (SConvs) following the 1×1 convolutions. This design not only reduces the parameter count but also improves the feature extraction capability.

B. Hierarchical Structure Mining

Constructing a hierarchical structure for HSI is a prerequisite for applying hierarchical classification. However, HSI lacks an intuitive hierarchical structure. Hence, it is essential to extract hierarchical features from the data to build such a structure. As discussed earlier, representing hierarchical structures in Euclidean space may result in distortion. To address this issue, we propose a HSM module based on hyperbolic space. This module constructs the hierarchical structure of the data by leveraging the interclass distances. Specifically, HSM can be divided into two parts: interclass distance calculation and distance-transformed hierarchical structure.

1) Interclass Distance Calculation: Let the HSI be $X \in R^{H\times W\times B}$, where H, W, B represent the height, width, and number of the bands, respectively, and $X_{\text{tr}} = \{\mathbf{x}_i\}_{i=1}^{N} \in X$ represent the training dataset, $Y_{\text{tr}} = \{y_i\}_{i=1}^{N}$ is the associated labels, where N represent the number of training data. According to the label information, the training dataset can be divided into similar constraints set S and dissimilar D constraints set as

$$S: \forall\left(\mathbf{x}_i, \mathbf{x}_j\right)|_{y_i=y_j} \in S \tag{12}$$

$$D: \forall\left(\mathbf{x}_i, \mathbf{x}_l\right)|_{y_i\neq y_l} \in D. \tag{13}$$

HSM first uses the backbone to extract features from the HSI, which can be formulized as

$$F_{\text{Euc}} = f(X) \tag{14}$$

where F_{Euc} is the output feature, and $f(\cdot)$ represents the backbone. Then, HSM map the output features $F_{\text{Euc}} \in R^{H\times W\times B^*}$ from Euclidean space to Lorentz model space, obtaining the output features $F_{\text{Hyp}} = \{f_i\}_{i=1}^{H\times W} \in H^{H\times W\times C}$ by (10), which express the hierarchical features with minimal distortion. Furthermore, to enhance the discriminative power of the features space, a triplet loss is applied to constrain the backbone

$$\text{Loss}_{\text{tri}} = \max\left(d_{H_\kappa^n}\left(x_i, x_j\right) - d_{H_\kappa^n}\left(x_i, x_j\right) + \text{margin}, 0\right) \tag{15}$$

where $d_{H_\kappa^n}(x_i, x_j)$ represent geodesic distance between x_i and x_j in the Lorentz model space. Utilizing the label information

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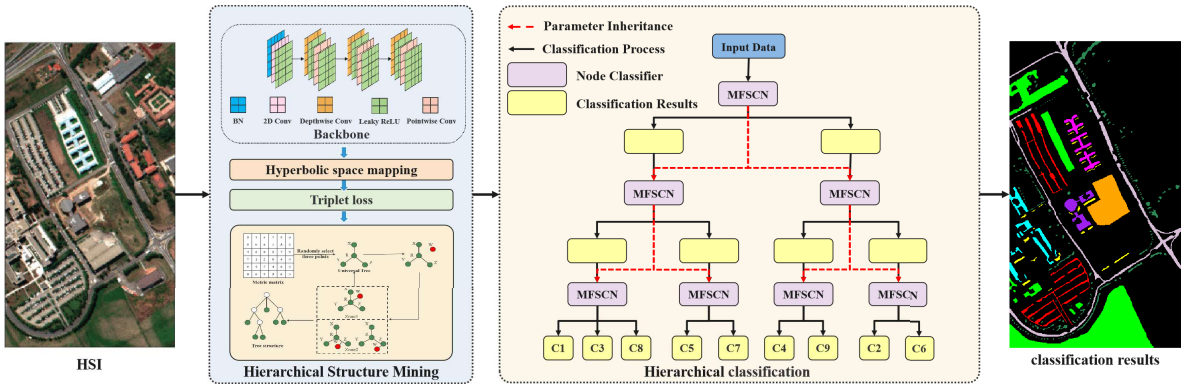


Fig. 3. Architecture of the proposed MFS-HiNet framework. First, the backbone is used to extract features from the HSI, which are then mapped into hyperbolic space to explore hierarchical representations and automatically construct the hierarchical structure. Based on the hierarchical structure, a hierarchical classification framework is designed, where a MFSCN is trained for each node, and a PIS is utilized to transfer model parameters from parent nodes to child nodes. Finally, the classification outcomes of the leaf nodes are aggregated to derive the final classification decision.

of the training samples, the mean feature vectors μ_c for each class within the training set can be computed

$$\mu_c = \frac{1}{|F_c|} \sum_{f_i \in F_c} f_i \tag{16}$$

where $F_c = \{f_i | y_i = c\}$ is the feature set corresponding to all samples in class c. By utilizing mean feature vectors μ_c, the interclass mean distance matrix $\mathbf{D}$ for the training samples can be computed.

2) Distance-Transformed Hierarchical Structure: In the previous stage, we obtained the interclass average matrix $\mathbf{D}$ of the training set samples. Using the method proposed by Sonthalia and Gilbert [53], we can transform $\mathbf{D}$ into a hierarchical structure T. The specific steps are as follows.

First, three classes c_x, c_y, c_z are randomly selected as the initial nodes, and the initial Steiner node is introduced to construct the initial hierarchical structure. The initial Steiner node r is obtained by computing the weighted average of the features of nodes c_x, c_y, and c_z.

Then, for the fourth category c_w, we calculate the Gromov product with categories c_x, c_y, and c_z respectively. The Gromov product is computed using the following formula:

$$\left(c_x, c_y\right)_{c_w} = \frac{1}{2}\left(\mathbf{D}_{(c_w,c_x)} + \mathbf{D}_{(c_w,c_y)} - \mathbf{D}_{(c_x,c_y)}\right). \tag{17}$$

Based on the Gromov product, the c_w can be assigned to one of seven distinct regions.

Situation 1: If c_w satisfies the following conditions:

$$\begin{cases} \max\limits_{i,j\in\{x,y,z\},i\neq j} \left|\left(c_i, c_j\right)_{c_w} - \mu\right| < \text{tol} \\ \max\limits_{i,j\in\{x,y,z\},i\neq j} \left(c_i, c_j\right)_{c_w} < \text{tol} \end{cases}$$

where tol represents the threshold, μ represents the mean of $(c_x, c_y)_{c_w}$, $(c_x, c_z)_{c_w}$, and $(c_y, c_z)_{c_w}$. The c_w is classified into the R1 region, indicating that the relationship between c_x, c_y, and c_z is very close. As a result, c_w is used to replace the Steiner node r.

Situation 2: If c_w satisfies the following conditions:

$$\begin{cases} \left|\left(c_x, c_y\right)_{c_w} - \max\limits_{i,j\in\{x,y,z\},i\neq j} \left(c_i, c_j\right)_{c_w}\right| < \text{tol} \\ \left|\left(c_x, c_y\right)_{c_w} - \max\limits_{j\in\{y,z\}} \left(c_x, c_j\right)_{c_w}\right| < \text{tol}. \end{cases}$$

The c_w is classified into the X1 region, indicating that c_w is closest to node c_x while also maintaining relatively close proximity to nodes c_y, and c_z, and therefore, during the construction of the hierarchical structure, c_w will be directly connected to node c_x. Similarly, this applies to the Y1 and Z1 regions as well.

Situation 3: If c_w satisfies the following conditions:

$$\begin{cases} \left|\left(c_x, c_y\right)_{c_w} - \max\limits_{i,j\in\{x,y,z\},i\neq j} \left(c_i, c_j\right)_{c_w}\right| < \text{tol} \\ \left|\left(c_x, c_y\right)_{c_w} - \max\limits_{j\in\{y,z\}} \left(c_x, c_j\right)_{c_w}\right| > \text{tol}. \end{cases}$$

The c_w is classified into the X2 region, indicating that c_w is closest to node c_x while maintaining a significantly larger distance from nodes c_y, and c_z, and therefore, during the construction of the hierarchical structure, c_w will be connected to node c_x through a newly introduced Steiner node. Similarly, this applies to the Y2 and Z2 regions as well.

Based on the aforementioned approach, each node to be incorporated into the hierarchical structure is gradually classified, and a hierarchical tree is recursively constructed.

C. Multifeature Space Classification Network

In the proposed MFS-HiNet, to enhance the capability of node classifiers in recognizing complex data and improving their sensitivity to subtle differences, we design the MFSCN to optimize the node classification task. This network employs a multibranch architecture, where each branch corresponds to a distinct feature space, enabling feature extraction from multiple perspectives. By integrating information from different feature spaces, this module effectively captures the diversity and hierarchical characteristics of the data, thereby improving classification accuracy and generalization ability. As illustrated in Fig. 4, the MFSCN consists of three branches, each corresponding to a different feature space: a local label feature branch in Euclidean space, a local label feature branch in hyperbolic space, and a global label feature branch in Euclidean space. Here, the local label represents the class label associated with the current node, while the global label encompasses the label information of the entire class hierarchy.

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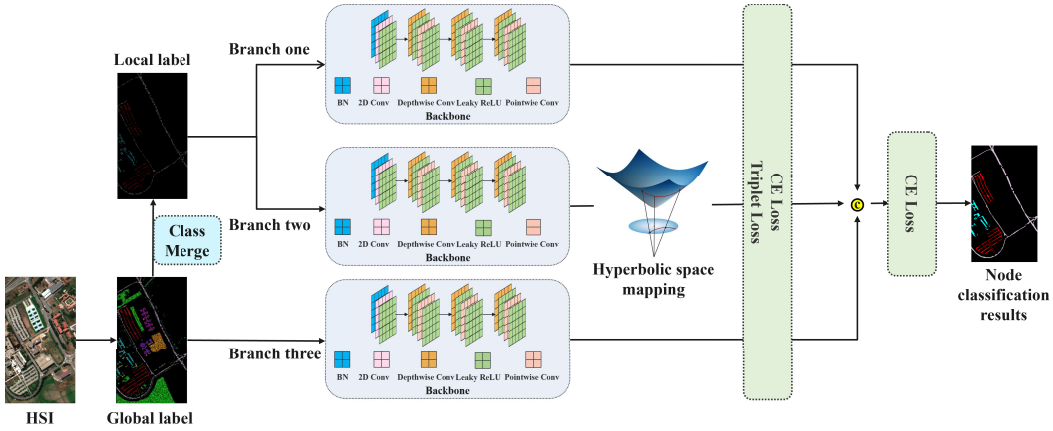


Fig. 4. Architecture of the proposed MFSCN. The MFSCN adopts a multibranch architecture consisting of three parallel branches, each employing an identical backbone structure for feature extraction. The three branches correspond to 1) local label features in Euclidean space; 2) local label features in hyperbolic space; and 3) global label features in Euclidean space. Here, the local label refers to the class associated with the current node, while the global label encompasses all leaf node categories, providing fine-grained semantic information. Each branch is supervised using both cross-entropy and triplet loss to enhance feature discriminability. Finally, features extracted from all branches are fused and further optimized with a unified cross-entropy to improve overall classification performance.

1) Branch One (Local Label Feature Branch in Euclidean Space): For this branch, the feature map $F_{\text{Euc}}^{\text{Loc}}$ is first extracted through the backbone, and a triplet loss $\text{TriLoss}_{\text{Euc}}^{\text{Loc}}$ based on the Euclidean distance is applied to $F_{\text{Euc}}^{\text{Loc}}$ to enhance the discriminative capability of the feature space. Subsequently, the extracted features are fed into a multilayer perceptron (MLP) to generate the classification probability map, which is further optimized by computing the cross-entropy loss $\text{CELoss}_{\text{Euc}}^{\text{Loc}}$ based on the class label of the corresponding node.

2) Branch Two (Local Label Feature Branch in Hyperbolic Space): Branch two and branch one use the same local labels for loss calculation and optimization, but they differ in geometric spaces. Branch two is based on hyperbolic space, where the features extracted by the backbone are mapped to hyperbolic space through (10) to obtain the feature representation $F_{\text{Hyp}}^{\text{Loc}}$. Subsequently, triplet loss $\text{TriLoss}_{\text{Hpy}}^{\text{Loc}}$ and cross-entropy loss $\text{CELoss}_{\text{Hpy}}^{\text{Loc}}$ are calculated to optimize this branch.

3) Branch Three (Global Label Feature Branch in Euclidean Space): Branch three, like branch one, performs computations in Euclidean space. However, the key difference lies in the utilization of global labels for loss calculation. Branch three produces the feature map $F_{\text{Euc}}^{\text{Glo}}$, and both triplet loss $\text{TriLoss}_{\text{Euc}}^{\text{Glo}}$ and cross-entropy loss $\text{CELoss}_{\text{Euc}}^{\text{Glo}}$ are applied to optimize the feature representation based on the global labels.

Finally, the features extracted from the three branches are fused to obtain the feature $F_{\text{fina}} = \text{concat}(F_{\text{Euc}}^{\text{Loc}}, F_{\text{Hyp}}^{\text{Loc}}, F_{\text{Euc}}^{\text{Glo}})$. Subsequently, the cross-entropy loss $\text{CELoss}_{\text{fina}}$ is calculated based on local labels, yielding the final classification result of the node. The overall loss function is formulated as follows:

$$L = \text{CELoss}_{\text{fina}} + \text{CELoss}_{\text{Euc}}^{\text{Loc}} + \text{CELoss}_{\text{Hpy}}^{\text{Loc}} + \text{CELoss}_{\text{Euc}}^{\text{Glo}} + \text{TriLoss}_{\text{Euc}}^{\text{Loc}} + \text{TriLoss}_{\text{Hpy}}^{\text{Loc}} + \text{TriLoss}_{\text{Euc}}^{\text{Glo}}. \quad (18)$$

D. Parameter Inheritance Strategy

In MFS-HiNet, node-by-node classification is employed to achieve HSI hierarchical classification. However, this approach typically trains classifiers independently at each hierarchical level, disregarding the inherent relationships between different levels.

To address this issue, we design the PIS, which takes advantage of the intrinsic correlations between classifiers at different levels. This strategy transfers the model parameters from parent nodes to child nodes, guiding the learning of child node classifiers, thereby enhancing their feature extraction capabilities. Specifically, the strategy is divided into three stages: 1) training of the parent node classifier; 2) parameter weighted inheritance; and 3) training of the child node classifier.

1) Training of the Parent Node Classifier: The classifier for the node Q can be obtained by training with the corresponding label of this node, and the model's corresponding parameters are denoted as θ_Q.

2) Parameter Weighted Inheritance: For the child node q of parent node Q, the initial parameters of its corresponding classifier are denoted as

$$\theta_q = \alpha\theta_Q + (1-\alpha)\,\theta_q^{\text{init}} \quad (19)$$

where α is a hyperparameter that controls the degree of parameter inheritance, allowing the child node classifier to both inherit useful information and adapt to new data distributions. In addition, θ_q^{init} is randomly initialized to prevent the parent node classifier from excessively constraining the lower-level task.

3) Training of the Child Node Classifier: Building upon the inherited parameters θ_q, the classifier of node q is subsequently trained using the corresponding labels to adapt to a more fine-grained classification task.

IV. Experiments

In this section, the experimental results of MFS-HiNet on five real-world datasets at different scales are first evaluated. Subsequently, a detailed report and analysis of the results are presented.

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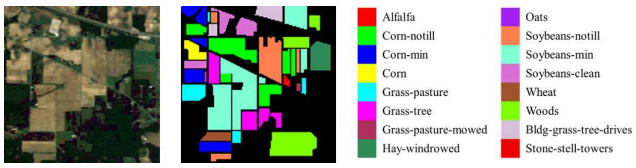


Fig. 5. Pseudocolor image and ground-truth for Indian Pines dataset.

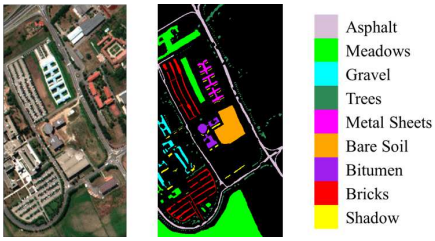


Fig. 6. Pseudocolor image and ground-truth for Pavia University dataset.

A. Dataset Description

In this article, five real-world HSI datasets, including Indian Pines, Pavia University, WHU-Hi-HongHu, WHU-Hi-HanChuan, and Houston 2013 datasets, are utilized to evaluate the performance of MFS-HiNet.

1) Indian Pines Dataset: Indian Pines dataset was captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over Northwestern Indiana, with a spatial resolution of 20 m and an image size of 145×145 pixels, comprising 16 distinct classes. The dataset covers a spectral range of 0.4–2.5 μm, consisting of 220 consecutive bands. After removing 20 bands affected by water absorption and noise, 200 bands remain available for analysis. Fig. 5 illustrates the corresponding false-color image and ground-truth map, respectively.

2) Pavia University Dataset: Pavia University dataset was captured by the reflective optics system imaging spectrometer (ROSIS) sensor in the University of Pavia, Italy, with a spatial resolution of 3.7 m and an image size of 610×340 pixels, comprising nine distinct classes. The dataset covers a spectral range of 0.43–0.86 μm, consisting of 103 consecutive bands. Fig. 6 illustrates the corresponding false-color image and ground-truth map, respectively.

3) Whu-Hi-HongHu Dataset: The WHU-Hi-HongHu dataset was captured in HongHu City, China, using a DJI Matrice 600 Pro UAV platform equipped with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor. The dataset has a spatial resolution of 0.046 m and an image size of 940×475 pixels, comprising 16 distinct classes. It covers a spectral range of 0.4–1 μm, consisting of 270 consecutive bands. Additionally, Fig. 7 illustrates the corresponding false-color image and ground-truth map, respectively.

4) Whu-Hi-HanChuan Dataset: The WHU-Hi-HanChuan dataset was captured in Hanchuan City, China, using a Leica Aibot X6 UAV V1 platform equipped with a 17-mm focal length Headwall Nano Hyperspec imaging sensor. The dataset has a spatial resolution of 0.109 m and an image size of 1217×303 pixels, comprising 16 distinct classes. It covers a spectral range of 0.4–1 μm, consisting of 274 consecutive bands. Fig. 8 illustrates the corresponding false-color image and ground-truth map, respectively.

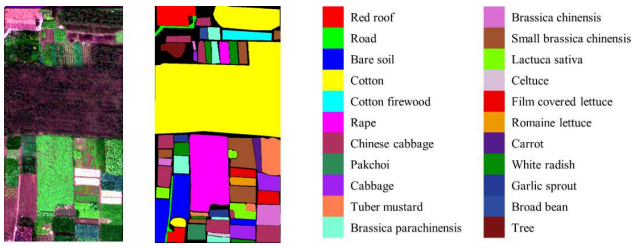


Fig. 7. Pseudocolor image and ground-truth for WHU-Hi-HongHu dataset.

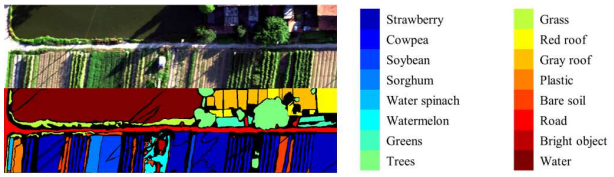


Fig. 8. Pseudocolor image and ground-truth for WHU-Hi-HanChuan dataset.

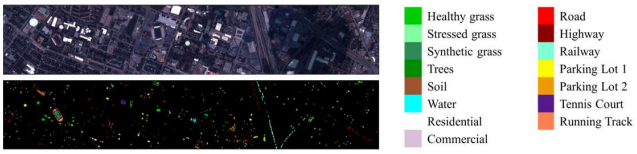


Fig. 9. Pseudocolor image and ground-truth map for Houston 2013 dataset.

5) Houston 2013 Dataset: Houston 2013 dataset was captured in Houston area, Texas, USA, by the National Center for Airborne Laser Mapping (NCALM) using the ITRES CASI-1500 sensor. The dataset has a spatial resolution of 2.5 m and an image size of 349×1905 pixels, comprising 15 distinct classes. It covers a spectral range of 380–1050 nm, consisting of 144 consecutive bands. Fig. 9 illustrates the corresponding false-color image and ground-truth map, respectively.

B. Experimental Setting

1) Implementation Details: MFS-HiNet is implemented using PyTorch, and all experiments are conducted on an A800 GPU. The classifiers at each node in MFS-HiNet are configured identically, with the learning rate set to $5e^{-4}$ and optimized using the Adam optimizer. The training iterations are set to 500, the SConv layers are set to 3, and the hyperparameter inheritance rate α is set to 0.5. In the experiments, training samples were randomly selected. Specifically, for Indian Pines dataset, ten samples per class were randomly chosen for training, 5 for validation, and the remaining samples were used for testing. For Pavia University dataset, 20 samples per class were selected for training, and 20 for validation, with the remainder used for testing. In the WHU-Hi-HongHu and WHU-Hi-HanChuan datasets, 50 samples per class were selected for training, 50 for validation, and the rest for testing. For Houston 2013 dataset, the official training and testing split provided by the competition organizers was adopted. All reported results are the average of five independent runs.

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TABLE I
CLASSIFICATION RESULT OF INDIAN PINES DATASET

Class	MSDN	SSFCN	Fully ContNet	CLOLN	SSGRN	Spectral Former	HSIC-FM	S^2VNet	CSIL	NSCKL	Graph GST	MFS-HiNet
1	55.08	88.89	82.41	81.46	100.00	84.26	92.59	27.92	99.07	100.00	100.00	**100.00**
2	52.60	19.56	50.73	67.76	53.87	36.84	34.39	61.48	42.38	**92.39**	80.62	62.84
3	38.72	29.96	57.76	49.78.	32.07	39.35	27.28	**77.88**	62.03	56.78	52.99	67.23
4	37.13	54.04	75.92	52.68	86.78	51.54	45.23	62.53	87.22	**96.35**	84.01	93.24
5	61.44	43.62	78.65	**95.84**	77.80	63.57	61.66	75.73	80.20	70.95	92.73	85.68
6	75.77	51.02	73.52	98.35	63.19	64.31	41.76	91.53	88.75	94.67	95.49	**98.46**
7	21.11	83.33	92.59	45.07	100.00	100.00	55.56	22.41	100.00	100.00	100.00	**100.00**
8	86.78	87.75	92.52	97.78	100.00	81.19	66.17	67.98	99.72	**100.00**	99.57	84.88
9	11.77	96.67	100.00	46.72	100.00	96.67	100.00	4.85	100.00	100.00	100.00	**100.00**
10	38.53	51.42	78.52	84.64	69.75	50.49	16.53	75.46	64.00	**87.04**	68.62	83.28
11	61.94	33.44	64.16	75.00	69.12	42.52	21.06	85.08	55.01	79.13	78.51	**89.22**
12	39.16	30.30	57.06	61.22	65.18	29.10	34.08	**86.60**	51.69	86.48	83.80	64.87
13	68.41	82.39	97.95	72.52	100.00	85.81	91.11	84.37	99.15	100.00	99.49	**100.00**
14	84.22	51.71	91.95	83.72	83.58	63.05	47.57	94.05	78.83	88.74	85.30	**97.12**
15	71.70	46.45	91.31	76.42	99.20	39.98	40.51	46.34	87.32	**99.73**	97.53	97.57
16	38.17	84.34	87.95	58.84	100.00	89.96	78.31	45.15	99.60	**100.00**	100.00	98.71
OA（%）	57.54	41.93	77.11	72.75	69.58	50.02	36.02	74.85	66.53	83.16	81.37	**83.74**
AA（%）	52.66	58.43	79.56	72.58	81.28	63.73	53.36	63.15	80.94	88.77	88.67	**88.94**
kappa×100	51.81	35.65	67.77	69.05	65.98	44.14	30.33	71.66	62.71	80.81	78.89	**81.27**

TABLE II
CLASSIFICATION RESULT OF PAVIA UNIVERSITY DATASET

Class	MSDN	SSFCN	Fully ContNet	CLOLN	SSGRN	Spectral Former	HSIC-FM	S^2VNet	CSIL	NSCKL	Graph GST	MFS-HiNet
1	91.48	65.00	67.52	98.47	68.40	62.66	67.56	94.02	75.37	81.98	89.44	**98.80**
2	88.88	80.74	76.88	96.41	81.05	76.77	85.05	99.16	88.5	**99.50**	94.72	91.13
3	46.08	80.55	89.07	92.50	87.59	77.22	59.15	66.64	91.44	**99.04**	92.15	97.71
4	89.81	94.03	80.95	91.08	82.88	93.65	81.89	76.86	90.24	87.94	92.24	**94.54**
5	85.69	93.69	98.16	98.23	100.00	100.00	97.31	97.29	100.00	100	100.00	**100.00**
6	47.77	79.96	86.34	91.33	94.09	70.71	60.20	94.78	99.18	99.08	96.14	**99.37**
7	68.50	85.65	95.47	99.12	94.35	86.26	74.07	92.60	97.86	99.92	96.87	**99.92**
8	75.76	67.06	91.23	83.36	72.25	62.76	74.89	**98.89**	82.77	81.24	87.46	93.95
9	82.61	97.27	96.04	87.94	87.59	97.88	93.67	80.22	92.56	84.79	98.27	**99.88**
OA（%）	76.48	78.88	80.31	93.11	81.45	75.37	77.28	92.86	88.23	94.02	93.45	**94.82**
AA（%）	75.18	82.28	86.85	91.94	85.35	80.88	77.09	88.94	90.88	92.61	94.15	**97.26**
kappa×100	69.52	72.99	75.01	90.88	76.36	68.56	70.36	90.66	84.76	92.08	91.38	**93.24**

2) Comparison Methods: In order to effectively validate the performance of the proposed algorithm, nine state-of-the-art algorithms were selected for comparison in the experimental section, including CNN-based methods: multi-scale dense network (MSDN) [54], SSFCN [33], FullyContNet [34], channel-layer-oriented lightweight spectral–spatial network (CLOLN) [55], spectral-spatial global graph reasoning (SSGRN) [56], Transformer-based methods: SpectralFormer [39], HSI classification full model (HSIC-FM) [38], S^2VNet [34], CSIL [40], and graph-based methods: normalized spectral clustering with kernel-based learning (NSCKL) [57], graph generative structure-aware transformer (GraphGST) [58]. All algorithms are implemented under the same experimental environment, using either the default settings from their open-source implementations or the hyperparameters recommended in the original publications to ensure a fair comparison.

C. Performance on the Five Datasets

In this section, we use five real-world datasets to quantitatively and qualitatively evaluate various classification methods to demonstrate the effectiveness of our approach.

1) Quantitative Evaluation: Tables I –V present the quantitative comparison between the proposed MFS-HiNet and state-of-the-art algorithms across five datasets. The compared methods include CNN-based methods, Transformer-based methods and Graph-based methods. The best classification results are highlighted in bold. Overall, MFS-HiNet achieves the highest performance in terms of OA, AA, and Kappa on four datasets; it only shows a 0.04% lower OA than the graph-based method NSCKL on Houston 2013 dataset, yet considering that NSCKL employs a semisupervised strategy, MFS-HiNet remains highly competitive. Specifically, the OA of MFS-HiNet on the five datasets are 83.74%, 94.82%, 93.33%, 95.06%, and 87.13%, respectively. Moreover, the superior AA performance of MFS-HiNet indicates that it not only achieves the best overall classification accuracy but also maintains balanced performance across classes under class-imbalanced conditions.

Further analysis of the experimental results reveals that CNN-based methods generally achieve superior classification performance on smaller datasets, such as Indian Pines and Pavia University datasets. For instance, the FullyContNet model achieves 77.11% OA on Indian Pines dataset,

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TABLE III
CLASSIFICATION RESULT OF WHU-HI-HANCHUAN DATASET

Class	MSDN	SSFCN	Fully ContNet	CLOLN	SSGRN	Spectral Former	HSIC-FM	S^{2}VNet	CSIL	NSCKL	Graph GST	MFS-HiNet
1	79.00	65.41	85.50	96.15	86.12	79.49	66.77	**97.86**	88.10	87.95	85.95	93.07
2	70.37	64.57	56.54	92.53	85.31	77.59	78.02	**93.28**	83.49	83.91	84.47	82.62
3	35.64	61.81	26.70	59.43	96.46	82.40	61.22	88.39	**96.77**	96.55	94.59	95.52
4	95.09	88.94	89.90	88.02	96.32	94.59	85.86	86.00	98.87	**99.32**	98.79	98.05
5	96.93	84.38	98.93	74.39	100.00	97.42	99.62	30.62	99.94	100.00	99.82	**100.00**
6	25.19	39.45	80.81	52.29	**98.68**	50.50	47.52	48.77	92.27	83.36	84.94	83.60
7	90.69	58.22	94.21	89.69	**97.77**	86.77	52.86	82.46	86.93	95.73	93.98	95.14
8	35.39	50.78	84.43	91.43	89.93	64.84	50.93	**96.78**	87.34	84.80	73.96	86.63
9	47.10	39.03	59.88	70.69	77.81	69.49	52.03	82.77	84.8	91.23	83.12	**94.06**
10	60.12	52.82	63.25	88.67	91.98	79.63	36.21	98.61	92.99	97.21	**99.67**	96.85
11	50.42	30.92	85.45	94.82	90.65	77.70	15.60	89.11	91.23	96.02	94.75	**97.24**
12	64.04	70.41	95.32	72.72	99.94	80.58	68.76	44.72	**99.99**	97.63	94.76	97.87
13	52.09	41.63	76.02	58.12	**90.20**	69.09	45.48	71.85	88.26	80.30	79.48	81.15
14	84.82	69.2	79.48	90.08	81.19	73.45	72.95	**91.28**	83.26	84.94	86.70	90.59
15	97.02	89.5	63.38	60.62	95.67	92.11	94.6	75.64	96.47	**99.63**	90.42	94.69
16	94.95	76.87	87.49	99.61	95.28	97.69	84.46	**99.98**	88.45	97.04	99.23	98.45
OA（%）	73.40	63.35	78.80	86.73	90.43	82.60	66.21	90.34	88.55	91.37	90.72	**93.33**
AA（%）	67.43	61.50	76.71	79.96	92.08	79.58	63.31	79.88	91.20	92.03	90.29	**92.84**
kappa×100	69.09	57.79	75.56	84.56	88.88	79.74	60.55	88.76	86.75	89.96	89.19	**92.21**

TABLE IV
CLASSIFICATION RESULT OF WHU-HI-HONGHU DATASET

Class	MSDN	SSFCN	Fully ContNet	CLOLN	SSGRN	Spectral Former	HSIC-FM	S^{2}VNet	CSIL	NSCKL	Graph GST	MFS-HiNet
1	73.82	80.74	78.29	95.20	96.73	94.51	66.32	**99.23**	93.05	96.33	98.32	97.22
2	89.69	78.79	60.18	80.97	79.86	91.06	85.91	81.65	95.90	96.04	94.10	**96.37**
3	64.26	67.7	85.21	98.00	91.26	79.22	75.81	**98.85**	93.96	87.25	89.79	92.08
4	90.11	81.41	73.83	99.48	85.84	93.12	73.81	**99.79**	91.47	92.55	96.57	97.48
5	66.95	60.53	92.27	68.49	**99.91**	89.18	68.98	53.80	98.87	96.92	96.09	97.19
6	88.81	76.67	68.22	96.32	69.17	89.66	82.36	**99.38**	93.98	97.94	92.16	95.74
7	34.58	48.47	26.87	91.40	62.22	64.63	44.84	**91.69**	79.68	83.91	85.51	90.41
8	34.49	34.17	81.17	54.66	**98.52**	62.40	58.40	61.55	98.43	94.06	92.45	91.40
9	87.52	76.45	56.46	89.31	89.46	95.01	74.54	97.61	94.27	97.93	98.38	**99.71**
10	73.75	58.24	66.86	84.71	58.49	65.30	79.34	**98.42**	93.03	88.60	89.67	92.91
11	41.96	39.76	51.95	66.22	68.51	57.98	38.83	**95.52**	85.52	93.69	88.23	90.05
12	63.45	60.67	67.14	69.82	88.85	68.95	59.34	80.75	**94.19**	85.74	79.93	84.94
13	68.56	50.10	56.42	80.22	74.81	65.17	54.05	81.85	**89.42**	78.14	81.30	84.68
14	64.67	70.85	70.70	78.09	84.84	79.28	73.51	94.73	**96.10**	92.94	92.17	93.61
15	90.44	78.99	93.03	44.52	100.00	95.55	92.40	80.46	99.96	99.26	98.31	**100.00**
16	94.65	81.44	66.56	97.24	98.98	90.75	87.12	**99.92**	97.51	99.28	94.37	98.20
17	89.00	84.21	92.71	88.21	**99.96**	90.44	90.02	97.06	99.27	95.44	97.06	98.23
18	90.21	65.99	82.70	77.60	97.12	90.05	67.97	78.29	98.58	**99.72**	97.12	98.32
19	65.36	52.88	66.13	70.47	77.47	87.60	60.16	93.57	91.04	93.74	90.64	**96.67**
20	88.86	63.27	93.02	67.00	99.33	94.83	74.19	81.97	**99.63**	99.21	94.35	98.75
21	59.57	66.82	99.53	66.89	100.00	91.94	86.25	47.08	**100.00**	100.00	99.29	99.66
22	86.60	79.98	93.03	56.61	**99.42**	93.65	68.44	79.20	98.20	98.85	97.34	95.00
OA（%）	78.55	71.62	67.12	87.89	82.19	85.33	70.47	93.97	91.86	92.05	93.06	**95.06**
AA（%）	73.06	66.51	73.81	78.25	87.31	83.20	71.03	86.02	94.64	93.98	92.87	**94.94**
kappa×100	73.45	65.67	61.34	84.83	78.21	81.76	64.73	92.43	89.89	90.11	91.2	**93.77**

significantly outperforming the transformer-based CSIL method, which only attains 66.53% OA. Similarly, on Pavia University dataset, the best CNN method achieves 93.11% OA, markedly higher than the 88.23% obtained by the transformer-based method. These results indicate that CNN methods excel at capturing local spatial details, making them more advantageous for small-scale datasets. However, as the dataset size increases (WHU-Hi-HanChuan and WHU-Hi-HongHu datasets), Transformer-based methods gradually exhibit superior performance. The CSIL method achieves OA values of 88.55% and 91.86% on the WHU-Hi-HanChuan and WHU-Hi-HongHu datasets, respectively, outperforming CNN methods, which obtain 86.73% and 87.89% OA. This is attributed to the self-attention mechanism in Transformer models, which effectively captures long-range dependencies and extracts global features across large spatial regions. In contrast, CNN methods primarily focus on local feature extraction, limiting their ability to model global spatial information comprehensively. Notably, despite its larger scale, Houston 2013 dataset exhibits highly fragmented and complex land-cover distributions, making local spatial information crucial for classification. Consequently, CNN-based methods achieve better performance on this dataset.

Graph-based methods can effectively capture local-to-global correlations among pixels in HSI, thereby enhancing feature discriminability and achieving excellent classification

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TABLE V
CLASSIFICATION RESULT OF HOUSTON 2013 DATASET

Class	MSDN	SSFCN	Fully ContNet	CLOLN	SSGRN	Spectral Former	HSIC-FM	S^2VNet	CSIL	NSCKL	Graph GST	MFS-HiNet
1	96.62	81.89	50.21	**97.38**	48.15	81.92	76.29	84.43	78.44	83.10	82.52	83.00
2	92.43	92.61	39.63	**98.51**	60.43	93.83	62.75	97.53	72.93	84.31	85.15	85.15
3	20.06	62.05	91.88	54.84	71.09	94.59	32.81	**100**	97.62	90.30	95.24	98.02
4	96.53	95.52	42.93	97.80	50.09	94.13	60.13	**100**	66.29	90.25	88.73	93.09
5	83.44	85.80	56.53	99.65	72.63	97.95	83.49	98.68	93.21	**100.00**	100.00	99.81
6	65.21	77.86	72.96	91.71	87.41	88.11	59.44	49.91	94.64	79.72	**100.00**	95.80
7	89.41	75.78	61.13	**89.36**	58.12	81.56	70.37	88.00	74.16	74.16	81.15	82.74
8	62.23	62.93	19.12	**93.59**	33.24	62.27	28.58	92.99	29.60	88.60	84.71	66.19
9	88.61	65.72	40.98	**90.59**	20.68	59.08	63.17	69.05	48.54	84.61	77.52	86.02
10	57.21	55.18	53.80	89.02	37.84	55.21	30.53	64.38	64.90	76.74	47.39	**95.66**
11	69.53	51.93	84.95	74.61	30.55	63.28	37.95	76.41	63.98	**87.19**	76.56	82.45
12	71.96	65.42	37.56	81.11	47.55	65.42	53.19	73.10	58.92	94.62	**95.58**	81.94
13	68.79	70.06	66.20	**92.79**	60.7	60.23	53.45	75.85	58.60	86.67	81.40	89.12
14	78.40	85.83	98.92	91.39	100	92.44	70.04	98.98	99.87	100.00	100.00	**100.00**
15	81.29	55.04	19.80	90.59	21.56	95.07	17.34	97.07	32.84	98.73	**100.00**	99.37
OA（%）	72.19	72.39	51.08	85.95	47.99	77.22	54.44	84.00	66.12	**87.19**	83.79	87.13
AA（%）	74.78	72.24	55.77	88.86	43.80	79.01	53.30	84.43	68.97	87.93	86.40	**89.22**
kappa×100	69.93	70.12	47.00	84.78	53.34	75.32	50.81	82.73	63.22	86.09	82.44	**86.09**

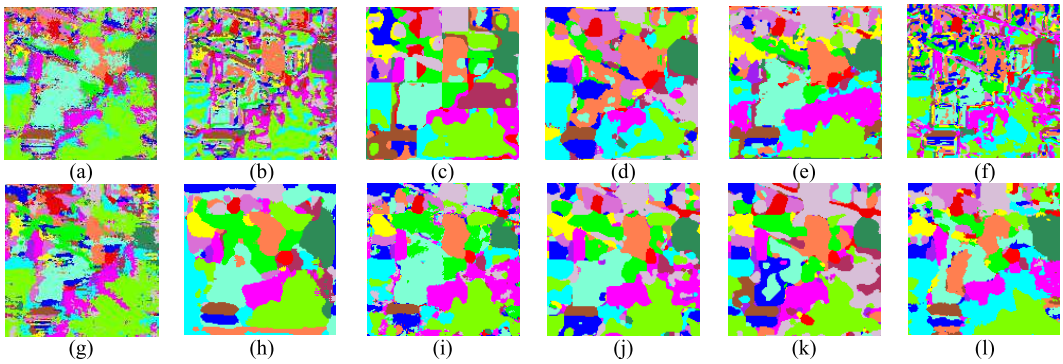


Fig. 10. Classification maps of different methods on Indian Pines dataset. (a) MSDN. (b) SSFCN. (c) FullyContNet. (d) CLOLN. (e) SSGRN. (f) SpectralFormer. (g) HSIC-FM. (h) S^2VNet. (i) CSIL. (j) NSCKL. (k) GraphGST. (l) MFS-HiNet.

performance. NSCKL constructs a pixel graph to aggregate local-to-global correlations and combines it with semi-supervised classification, achieving 86.16%, 94.02%, and 91.37% on Indian Pines, Pavia University, and WHU-Hi-HanChuan datasets, respectively, ranking as the second-best results. GraphGST, on the other hand, introduces graph structures and absolute positional encoding within a Transformer framework, integrating local neighborhood graphs and cross-view contrastive learning with a self-supervised strategy, and also achieves strong classification performance.

The proposed MFS-HiNet organically integrates global and local features through the hierarchical classification strategy. Specifically, the coarse-grained node classifiers extract global-scale discriminative information, while the fine-grained nodes progressively refine the classification to capture local features. The PIS further facilitates the transfer of global information from coarse-grained nodes to guide the feature learning of fine-grained nodes. This joint modeling of global and local features enables MFS-HiNet to achieve the best classification performance across datasets of varying scales.

2) Qualitative Evaluation: To provide a more intuitive comparison of the classification performance of different methods, the classification maps generated by each method are presented in Figs. 10–14. It can be observed that MSDN and SSFCN exhibit strip noise and salt-and-pepper noise across five datasets due to the limited feature extraction capability, which leads to the loss of contextual spatial information. FullyContNet and CLOLN suffer from severe oversmoothing effects, mainly caused by the insufficient extraction of spectral information. In contrast, Transformer-based methods mitigate noise to some extent but struggle to distinguish spectrally similar land cover types, resulting in frequent misclassifications. Graph-based methods further improve the overall smoothness of the classification results and reduce misclassifications to some extent, but they remain insufficient in distinguishing fine-grained categories. The proposed framework yields more refined and clearer classification maps, alleviating the oversmoothing phenomenon, significantly reducing noise points, and accurately distinguishing easily confused classes. For instance, our method effectively differentiates soybeans-notill and soybeans-min in Indian Pines dataset, meadows and bare soil in Pavia University dataset, and delineates the boundaries of roads and buildings in Houston 2013 dataset. These qualitative results further validate the effectiveness of the proposed framework from a visual perspective.

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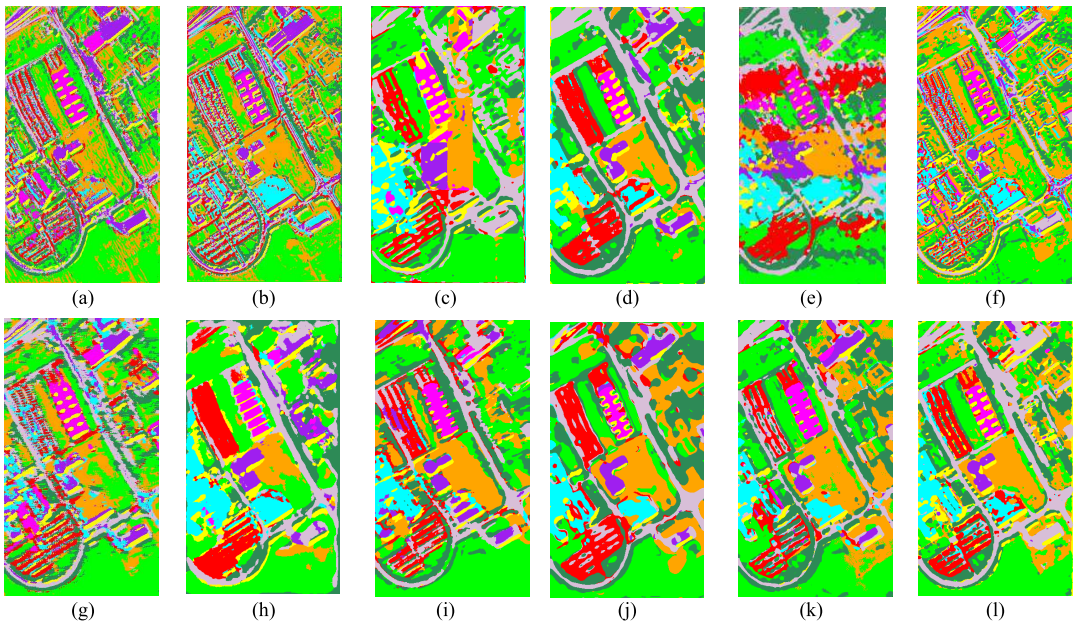


Fig. 11. Classification maps of different methods on Pavia University dataset. (a) MSDN. (b) SSFCN. (c) FullyContNet. (d) CLOLN. (e) SSGRN. (f) SpectralFormer. (g) HSIC-FM. (h) S2VNet. (i) CSIL. (j) NSCKL. (k) GraphGST. (l) MFS-HiNet.

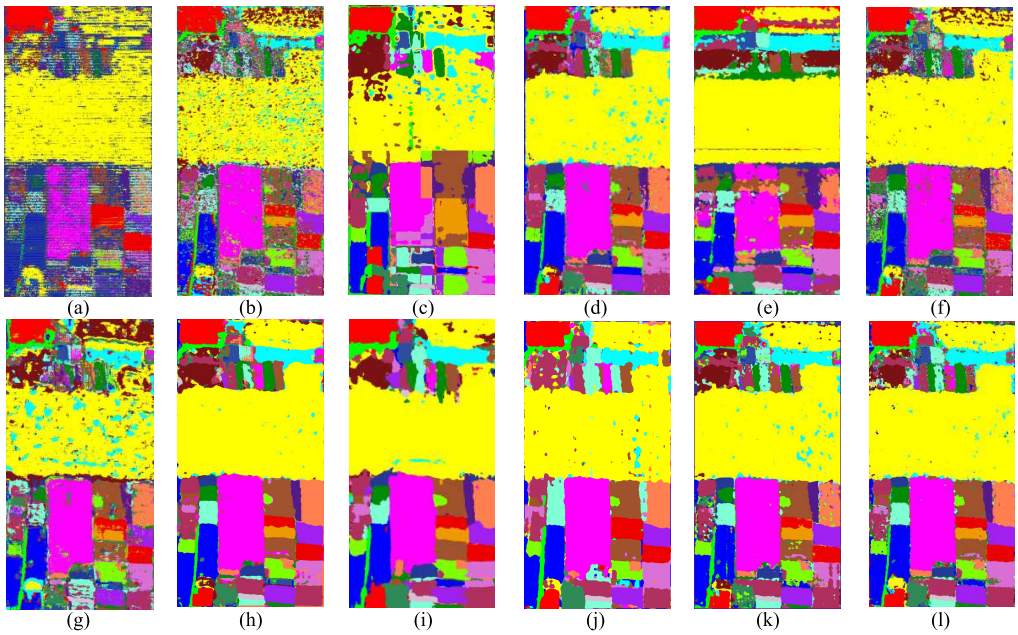


Fig. 12. Classification maps of different methods on the WHU-Hi-HongHu dataset. (a) MSDN. (b) SSFCN. (c) FullyContNet. (d) CLOLN. (e) SSGRN. (f) SpectralFormer. (g) HSIC-FM. (h) S^2VNet. (i) CSIL. (j) NSCKL. (k) GraphGST. (l) MFS-HiNet.

D. Parameter Analysis

In MFS-HiNet, there are two hyperparameters that require manual configuration: the number of SConv layers in the backbone and the inheritance ratio in the PIS. These parameters are critical to classification performance; therefore, we conducted an in-depth analysis to investigate their impact on the model's performance under different settings.

1) Number of SConv Layers: The number of SConv layers directly affects the model's feature extraction capability. As shown in Table VI, increasing the number of layers appropriately enables the extraction of deeper features, enhancing the ability to capture complex data patterns. OA gradually improves in the initial stage. When the number of layers is set to 3, the classification performance reaches its optimal level. However, as the number of layers continues to increase, the model's complexity rises, leading to a higher risk of overfitting and a decline in generalization ability, ultimately causing OA to decrease.

2) Inheritance Ratio in the PIS: This parameter controls the proportion of parameters inherited by the child node classifier from the parent node classifier, thereby reflecting the extent to which the child node inherits information from the parent

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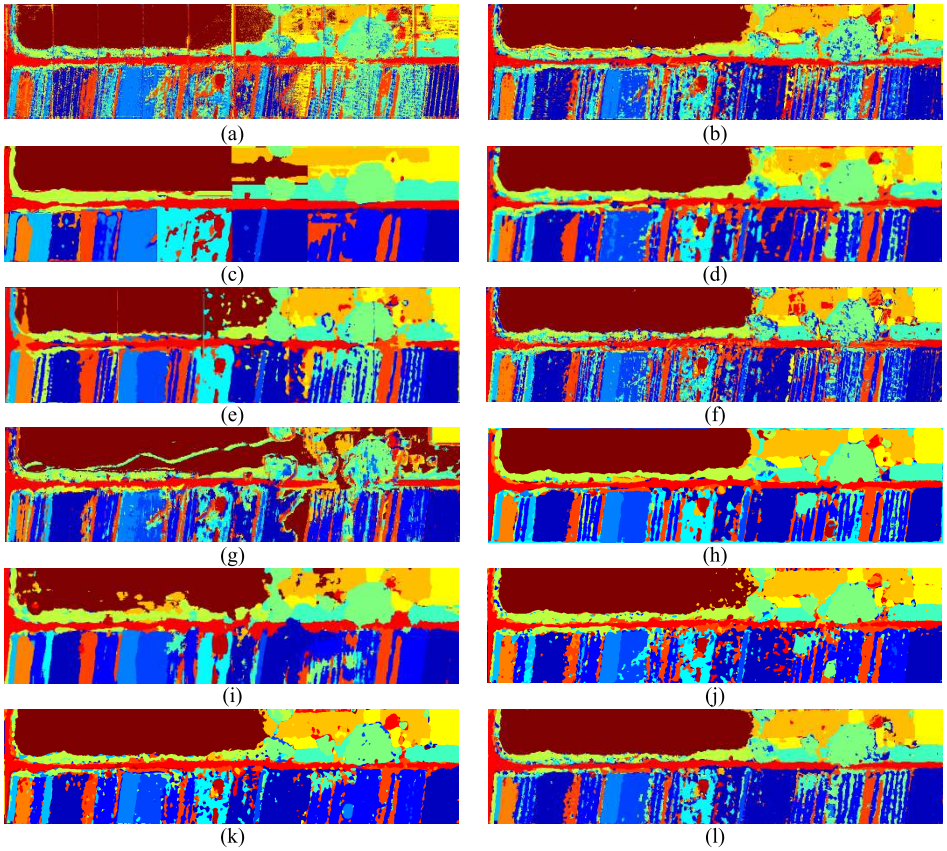


Fig. 13. Classification maps of different methods on the WHU-Hi-HanChuan dataset. (a) MSDN. (b) SSFCN. (c) FullyContNet. (d) CLOLN. (e) SSGRN. (f) SpectralFormer. (g) HSIC-FM. (h) S2VNet. (i) CSIL. (j) NSCKL. (k) GraphGST. (l) MFS-HiNet.

TABLE VI
INFLUENCE OF DIFFERENT SCONV LAYERS ON CLASSIFICATION RESULTS

SConv layers	Indian Pines	Pavia University	WHU-Hi-HanChuan	WHU-Hi-HongHu	Houston 2013
1	81.52	91.56	88.58	87.44	82.57
2	83.03	93.73	90.94	91.36	85.20
3	**83.74**	**94.82**	**93.33**	**95.06**	**87.14**
4	76.97	92.99	92.45	92.76	83.95
5	76.05	92.36	92.10	91.34	83.09

node. To investigate this, we analyzed the impact of different inheritance ratios on classifier performance, and the corresponding OA are presented in Fig. 15. As shown in the figure, the OA increases as the inheritance ratio rises, indicating that the parent node can provide valuable information for the child node classification task at the initial stage. When the inheritance ratio reaches 0.5, the OA achieves its highest, suggesting that at this point, the classifier effectively balances the information from the parent node, inheriting useful features while adapting to the new data distribution. However, when the inheritance ratio exceeds 0.5, the OA begins to decrease, likely due to the excessive constraint imposed by the parent node, which prevents the child node from effectively adapting to the new classification task. Moreover, it is noteworthy that when the inheritance ratio is set to 1, the OA of the classifier is significantly higher than that when the inheritance ratio is set to 0, which validates the importance of inheriting information from the parent node and further demonstrates the effectiveness of the hierarchical PIS.

Based on the above analysis, the number of SConv layers is set to 3, and the inheritance ratio is set to 0.5 in this article.

E. Ablation Experiment

In this section, we design a series of ablation experiments to evaluate the impact of each module in the MFS-HiNet framework on the classification results.

1) Impact of Hierarchical Structure: Hierarchical structure is a fundamental element of hierarchical classification, and its validity and accuracy directly influence classification performance. A well-constructed hierarchical structure can effectively organize the relationships between classes, allowing the classification model to fully leverage hierarchical information during training, thereby improving generalization and robustness. Therefore, this section explores the impact of constructing hierarchical structures in both Euclidean and hyperbolic spaces on classification results.

As shown in Table VII, the hierarchical structure built in hyperbolic space yields better results when applied to hierarchical classifiers across five datasets. This indicates that hyperbolic space can more accurately represent the hierarchical relationships between classes, enabling the classifier to better capture the organizational structure of the classes, thereby enhancing classification performance. In contrast, in Euclidean space, the ability to represent hierarchical structures

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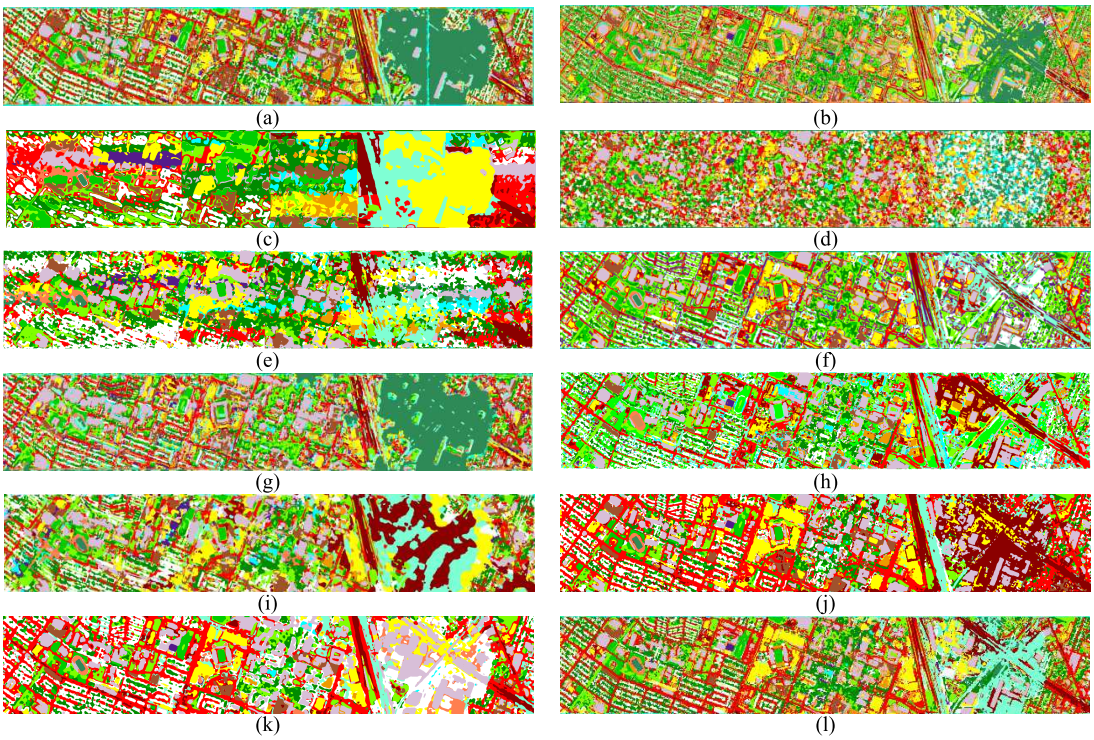


Fig. 14. Classification maps of different methods on Houston 2013 dataset. (a) MSDN. (b) SSFCN. (c) FullyContNet. (d) CLOLN. (e) SSGRN. (f) SpectralFormer. (g) HSIC-FM. (h) S2VNet. (i) CSIL. (j) NSCKL. (k) GraphGST. (l) MFS-HiNet.

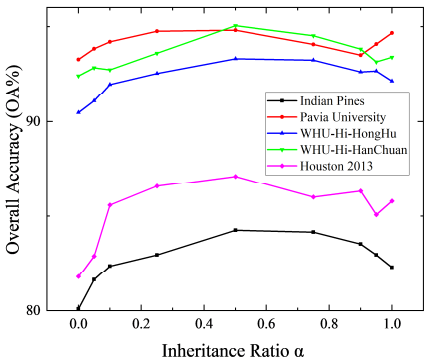


Fig. 15. Influence of different inheritance ratio α on classification results.

TABLE VII
IMPACT OF DIFFERENT HIERARCHICAL STRUCTURES ON CLASSIFICATION RESULTS

	Indian Pines	Pavia University	WHU-Hi-HanChuan	WHU-Hi-HongHu	Houston 2013
Euclidean space	81.27	92.35	92.01	91.56	84.76
Hyperbolic space	**84.26**	**94.82**	**93.33**	**95.06**	**87.13**

is limited, causing the relationships between categories to distort, which hampers the classifier's ability to fully utilize hierarchical information and ultimately impacts the classification outcome.

2) Impact of Different Feature Space: This section investigates the impact of different feature branches in the MFSCN on classification performance, using the Euclidean space-based local label branch as the baseline. Table VIII presents the classification results under different branch combinations.

TABLE VIII
INFLUENCE OF DIFFERENT FEATURE SPACES ON CLASSIFICATION RESULTS

	Indian Pines	Pavia University	WHU-Hi-HanChuan	WHU-Hi-HongHu	Houston 2013
branch one	77.66	91.78	90.07	94.06	83.74
+branch two	82.79	93.29	91.96	94.85	86.51
+branch three	**84.26**	**94.82**	**93.33**	**95.06**	**87.13**

The experimental results demonstrate that incorporating both the local label feature branch in hyperbolic space (branch two) and the global label feature branch in Euclidean space (branch three) effectively enhances overall classification performance. Specifically, the branch two enables more precise capture of hierarchical relationships between classes and extracts hierarchical information from hyperbolic space, compensating for the limitations of Euclidean space in representing hierarchical structures. This branch provides a more structured representation of class relationships, thereby improving classification accuracy and reducing misclassifications across hierarchical levels. Moreover, the branch three facilitates information sharing among different classification nodes, ensuring that the classifier leverages category features more comprehensively across multiple levels. This helps improve overall classification stability and consistency.

3) Impact of Different Modules: This section evaluates the influence of three modules within the MFS-HiNet framework

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TABLE IX
IMPACT OF DIFFERENT MODULES ON CLASSIFICATION RESULTS

Baseline	Hierarchical classification	MFSCN	PIS	Indian Pines	Pavia University	WHU-Hi-HanChuan	WHU-Hi-HongHu	Houston 2013
√				68.85	88.94	87.39	88.17	80.75
√	√			74.12	90.36	89.02	89.89	81.36
√	√	√		80.09	93.29	90.47	92.43	81.79
√	√		√	77.66	91.78	90.07	94.06	83.74
√	√	√	√	**84.26**	**94.82**	**93.33**	**95.06**	**87.13**

TABLE X
COMPARISON OF MODEL PARAMETERS (M), FLOPS (G) AND INFERENCE TIME (S) ACROSS FIVE DATASETS

Dateset	Method	MSDN	SSFCN	Fully ContNet	CLOLN	SSGRN	Spectral Former	HSIC-FM	S^2VNet	CSIL	NS CKL	Graph GST	MFS-HiNet
Indian Pines	Parameters (M)	1.58	0.11	1.37	0.05	3.05	0.34	17.69	0.23	170.20	-	2.92	0.02
	FLOPs(G)	0.38	14.24	9.45	0.01	14.34	4.57	103.21	4.24	37.19	-	0.01	0.56
	Inference time(s)	1.35	0.06	0.80	1.4	1.39	1.9	0.73	2.21	26.58	-	0.8	0.24
Pavia University	Parameters (M)	1.58	0.09	1.311	0.03	32.38	0.16	14.50	0.11	136.52	-	5.34	0.01
	FLOPs(G)	0.20	120.90	78.94	0.01	101.72	1.87	102.59	1.55	36.18	-	0.01	3.32
	Inference time(s)	3.78	0.17	1.25	3.1	5.12	5.8	35.67	4.95	96.14	-	3.9	0.31
WHU-Hi-HongHu	Parameters (M)	1.58	0.12	1.41	0.06	147.32	0.55	19.99	0.37	195.88	-	5.51	0.03
	FLOPs(G)	0.45	264.88	176.62	0.01	285.01	7.09	103.51	7.10	37.95	-	0.17	16.33
	Inference time(s)	447.91	0.79	1.57	91.9	6.23	104.9	881.27	114.6	523.36	-	34.0	0.68
WHU-Hi-HanChuan	Parameters (M)	1.59	0.12	1.41	0.06	96.81	0.54	20.12	0.38	194.49	-	6.21	0.03
	FLOPs(G)	0.52	319.50	212.38	0.01	226.92	7.25	103.70	7.29	37.91	-	0.07	13.34
	Inference time(s)	375.26	0.65	1.55	81.1	5.78	88.5	758.61	78.0	479.35	-	54.5	0.48
Houston 2013	Parameters (M)	1.58	0.10	1.33	0.04	316.05	0.22	15.85	0.15	150.76	-	4.23	0.02
	FLOPs(G)	0.27	410.13	267.19	0.01	792.29	2.89	102.85	2.51	36.61	-	0.07	14.42
	Inference time(s)	47.91	0.58	1.23	3.2	4.59	2.1	393.49	1.77	176.83	-	1.8	0.36

on classification results, including the hierarchical classification strategy, MFSCN, and PIS. We employ a model that utilizes the backbone as the classifier and adopts a flat classification strategy as the baseline. Table IX shows the classification results under various module combinations.

As can be observed from Table IX, the hierarchical classification strategy, MFSCN, and PIS all significantly enhance the classification accuracy across all datasets. Compared to the flat classification strategy, mining the hierarchical structure within the data and adopting the hierarchical classification strategy effectively improves classification accuracy. When only hierarchical classification is employed, the OA of the five datasets increases by 5.27%, 1.42%, 1.63%, 1.71%, and 0.61%, respectively. Replacing the backbone with the MFSCN leads to a substantial improvement in OA. This is attributed to the integration of information from different feature spaces, enabling the model to more comprehensively capture the diversity, hierarchy, and complex relationships within the data. On the five datasets, the OA increases by 11.24%, 4.35%, 3.08%, 4.26%, and 1.04%, respectively. Additionally, with the assistance of the PIS, lower-level nodes can inherit parameters from upper-level nodes, facilitating information sharing across different levels and further improving the classification accuracy of each node. Experimental results demonstrate that PIS effectively enhances classification accuracy, whether using the backbone or the multifeature space local classifier. Overall, when hierarchical classification, multifeature space local classifier, and PIS are employed simultaneously, the model achieves the highest accuracy on all datasets. Compared to the baseline model, the OA of the five datasets increases by 15.41%, 5.88%, 6.60%, 6.89%, and 6.38%, respectively. These results validate the effectiveness of the various modules we have proposed.

F. Computation Cost Analysis

We conducted a comprehensive analysis of the computational cost associated with the proposed method and several competing approaches, focusing on three key metrics: the number of model parameters, floating-point operations (FLOPs), and inference time. Since NSCKL is a traditional machine learning algorithm rather than a deep learning method, its computational characteristics and model structure differ fundamentally, and thus it was excluded from this computational cost comparison. The results are summarized in Table X.

The MFS-HiNet demonstrates a clear advantage in terms of model size, achieving the smallest number of parameters among all compared methods. This is primarily attributed to the lightweight design of the MFS-HiNet backbone, which is composed solely of 1×1 convolutions and SConvs. Although MFS-HiNet does not achieve the lowest FLOPs, the difference compared to the most efficient model is marginal and remains within an acceptable range. Regarding inference time, MFS-HiNet consists of multiple node-specific classifiers, and the reported inference time reflects the cumulative runtime of all these classifiers. Despite utilizing the LCPN design, our

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method achieves the fastest inference time. This benefit stems from the design choice of inputting the entire image into the model at once, thereby eliminating the need for patchwise or sliding window operations and substantially reducing redundant computations.

In summary, MFS-HiNet achieves superior classification performance while maintaining a low parameter count and high inference efficiency, effectively reducing computational resource requirements and significantly enhancing inference speed and deployment flexibility.

V. Conclusion

This article proposes a MFS-HiNet designed to enhance the discrimination of fine-grained categories. By explicitly modeling hierarchical relationships and guiding the classification process in a top-down manner, the framework strengthens the ability to distinguish subtle interclass differences. Specifically, the HSC module automatically constructs the hierarchical structure of hyperspectral data in hyperbolic space, revealing interclass relationships with low distortion and addressing the lack of predefined hierarchical structures. Based on this structure, the MFSCN node classifier integrates features from multiple feature spaces to achieve cross-space information complementarity, thereby enhancing discriminative power, while the PIS module guides the learning of child nodes through parent-node parameters, promoting interlevel information sharing and further improving classification accuracy. Experimental results on five hyperspectral datasets demonstrate that MFS-HiNet outperforms existing methods, validating the effectiveness of hierarchical classification in hyperspectral tasks and providing a novel solution for the field. Despite its superior classification performance, MFS-HiNet relies entirely on spectral information to construct the hierarchy, making it susceptible to variations in data distribution, and the hierarchy derived from spectral similarity may differ from human-perceived semantic hierarchies. Future work will focus on integrating spectral and semantic information to construct more robust, interpretable, and semantically consistent hierarchies, thereby further enhancing the model's generalization ability and classification performance.

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